# EDNS 491 & 492: Capstone Design I & II

## Student Guide

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**How to use this guide:** This document is your sprint-by-sprint companion for Capstone Design. It is *not* a replacement for the Course Syllabus, the Course Text, or Canvas; it is a navigation aid that tells you where to focus your energy each sprint. For each sprint, start by reading that sprint's section and pay special attention to any box labeled "Common Pitfall" or "Pro Tip" (those address the most frequent issues). Read the relevant section at the *start* of each sprint, not the end.

**Three documents work together**: (1) this Student Guide (sprint-level navigation), (2) the Course Text (deep reference, 22 chapters across 5 parts), and (3) Canvas (assignments, rubrics, deadlines). The Course Syllabus is the authoritative source for grading and policies.

**A note on numbering**: section numbers in this Student Guide (e.g., "Section 3") refer to sections *within this guide*, not to chapters in the Course Text. When referencing the Course Text, this guide always uses the format "Course Text Ch. X" to avoid confusion.

## Contents

**1 The Capstone Sprint: A Typical Week** **4**

**2 Course Philosophy and the Three-Document Ecosystem** **4**
2.1 How SDI and SDII Fit Together . . . 5
2.2 Your Three Documents . . . 5

**3 The Most Common Mistake at Each Sprint** **5**

**4 SDI Sprint 0: Project Expo, Team Formation, Client Kickoff** **6**
4.1 Deliverables . . . 7

**5 SDI Sprint 1: The Project Plan, Scrum Training, Client Meeting Prep** **7**
5.1 Deliverables . . . 7

**6 SDI Sprint 2: Concept Generation, Trade Study, Career Day** **7**
6.1 Deliverables . . . 8

**7 SDI Sprint 3: Concept Development, Client Approval of Approach** **8**
7.1 Deliverables . . . 8

**8 SDI Sprint 4: Functional Prototypes / Site Plans, Ethics Assignment** **9**
8.1 Deliverables . . . 9

**9 SDI Sprint 5: PDR Preparation, Mock PDR, PDR Execution** **9**
9.1 Deliverables . . . 10

**10 SDI Sprint 6: Design Refinement, Next Steps, SDII Handoff** **10**
10.1 Deliverables . . . 10
10.2 Preparing for SDII . . . 11

**11 SDII Sprint 7: Re-engagement, Goal Setting, Hazard Assessment** **11**
11.1 Deliverables . . . 11

**12 SDII Sprint 8: CDR Report, Eng Calc Package, CDR Execution** **11**
12.1 Deliverables . . . 12

**13 SDII Sprints 9–11: Build/Test/Iterate, Broader Impacts Essay** **12**
13.1 Deliverables . . . 12

**14 SDII Sprints 12–13: FDR Preparation, System V&V, Final Testing** **13**
14.1 Deliverables . . . 13

**15 SDII Sprint 14: FDR, Showcase, Documentation, Final Checkout** **13**
15.1 Final Deliverables . . . 14

**16 Design Review Preparation Checklist** **14**

**17 What Your PA Expects** **15**

**18 The Feedback Loop Survival Guide** **15**

**19 CFO and Purchasing Survival Guide** **16**

**20 Travel Planning Quick Reference** **16**

**21 Resume and Career Day Preparation** **17**

**22 Module Checklist: Sprint-to-Chapter Mapping** **17**

**23 Quick Reference: Capstone Abbreviations** **18**

**24 Emergency Contacts and Key Resources** **18**

**A Assignment and Rubric Reference** **19**
A.1 EDNS 491 — Senior Design I Assignments . . . 19
A.1.1 Sprint Cycle Assignments (Sprints 0–5) . . . 19
A.1.2 Project Kickoff & Client Meeting . . . 20
A.1.3 Project Plan (Statement of Work) . . . 20
A.1.4 Preliminary Design Report (PD Report) . . . 20
A.1.5 Preliminary Design Review (Presentation) . . . 20
A.1.6 Ethics Assignment . . . 21
A.1.7 Engineering Standards . . . 21
A.1.8 Project Goal Setting & Goal Attainment . . . 21
A.1.9 Individual Performance Evaluations . . . 22

A.1.10 Next Steps Email . . . 22
A.2 EDNS 492 — Senior Design II Assignments . . . 22
A.2.1 Sprint Cycle Assignments (Sprints 7–14) . . . 22
A.2.2 Re-engagement Email & Activity . . . 22
A.2.3 Critical Design Report (CDR) . . . 23
A.2.4 Critical Design Review (Presentation) . . . 23
A.2.5 Final Design Report (FDR) . . . 23
A.2.6 Final Design Review (Presentation) . . . 23
A.2.7 Broader Impacts Essay . . . 23
A.2.8 Design Showcase (Poster) . . . 24
A.2.9 Project Synopsis . . . 24
A.2.10 Digital Design Archive . . . 24
A.2.11 "What We Wish We Would Have Known" Video . . . 24
A.2.12 Manufacturing Readiness Review . . . 24
A.3 Rubric Reference . . . 24
A.3.1 Preliminary Design Report Rubric . . . 24
A.3.2 Preliminary Design Review Presentation Rubric . . . 25
A.3.3 Critical Design Report Rubric . . . 26
A.3.4 Critical Design Review Presentation Rubric . . . 26
A.3.5 Final Design Report Rubric . . . 27
A.3.6 Final Design Review Presentation Rubric . . . 28
A.3.7 Ethics Assignment Rubric . . . 28
A.3.8 Broader Impacts Essay Rubric . . . 28
A.3.9 Project Plan (Statement of Work) Rubric . . . 29
A.3.10 Design Showcase Poster Rubric . . . 30
A.3.11 Individual Performance Evaluation Rubric . . . 30
A.3.12 Next Steps Email Rubric (SDI) . . . 31
A.4 Grade Weights . . . 32
A.4.1 EDNS 491 — Senior Design I . . . 32
A.4.2 EDNS 492 — Senior Design II . . . 32

# 1 The Capstone Sprint: A Typical Week

**This course meets formally on Tuesdays and Thursdays only** (lectures and recitations). The Monday, Wednesday, and Friday activities shown below are *self-directed* team work sessions that you are responsible for organizing outside of class time. Capstone requires significant independent effort beyond scheduled meetings. Teams that rely solely on Tuesday/Thursday class time will fall behind.

**Attendance at every lecture and recitation is mandatory.** Key deliverable instructions, client feedback, design review logistics, and PA assessments occur during these sessions. Material covered in lecture and recitation cannot always be recovered from slides alone. Missing a session means missing information your team needs to stay on track.

Your team's weekly rhythm should follow this general pattern (adapt to your specific schedule):

| Day | Activities |
|---|---|
| Monday | *Self-directed*: Team stand-up (15 min). Update task board. Identify blockers. |
| Tuesday | **Scheduled**: Lecture / Recitation / Lab session. Design, analysis, or fabrication work. |
| Wednesday | *Self-directed*: Work session. PA check-in (show progress, surface issues). |
| Thursday | **Scheduled**: Lecture / Recitation / Lab session. Send client status email (biweekly). |
| Friday | *Self-directed*: Work session. Documentation. End of Week 2: Sprint Review + Retrospective. |

The three non-negotiable weekly touchpoints: (1) the stand-up (team coordination), (2) the PA check-in (accountability), and (3) the client communication (professionalism). Everything else flexes around your project needs. Plan for 6–10 hours per week of team work beyond scheduled class time.

# 2 Course Philosophy and the Three-Document Ecosystem

Capstone Design is not another course; it is a professional practice experience compressed into two semesters. You will define problems, manage a team, communicate with a real client, design and build (or model) a system, verify it against requirements, and present your results publicly at the Design Showcase. The skills you develop here (engineering judgment, project management, client communication, teamwork) are the skills employers value most.

**What level of independence is expected?**

Capstone demands self-motivation. Unlike lecture-based courses, your progress depends on work you organize and complete *outside* of scheduled Tuesday/Thursday class sessions. Your PA will not chase you for deliverables. Your client will not remind you of deadlines. You and your teammates are responsible for holding each other accountable, scheduling your own work sessions, and maintaining momentum between class meetings. Treat this course the way you would treat your first engineering job: your supervisor sets the objectives, but you manage your own time to meet them.

### 2.1 How SDI and SDII Fit Together

**SDI** (EDNS 491) is the **definition semester**: you go from project assignment through requirements, concept selection, and Preliminary Design Review. Sprints 0–6 cover the left side of the V-Model.

**SDII** (EDNS 492) is the **execution semester**: you re-engage with the client, complete the Critical Design Review, build/model your system, test it, and present at the Final Design Review and Design Showcase. Sprints 7–14 cover the bottom and right side of the V-Model.

### 2.2 Your Three Documents

- **This Student Guide**: tells you what to focus on each sprint.
- **The Course Text**: provides the engineering methodology, process guidance, and technical reference. Organized by topic in 22 chapters across 5 parts.
- **Canvas**: assignment rubrics, submission portals, professional development modules, scheduling, and the Labriola InnoHub enrollment site.

## 3 The Most Common Mistake at Each Sprint

| Sprint | Most Common Mistake | Prevention |
|---|---|---|
| 0 | Starting to design before understanding the problem | Spend Sprint 0 listening, not solving. Ask questions, take notes. |
| 1 | SOW with no "Out of Scope" section | Explicitly list what you will NOT do. Get client agreement. |
| 2 | Generating only one concept | Use brainstorming, SCAMPER, morphological chart. Generate $\geq$15 ideas. |
| 3 | Decision matrix rigged to support a predetermined choice | Add sensitivity analysis. Vary weights by $\pm$20%. |
| 4 | PoC that does not retire the key technical risk | Identify the risk first, then design the PoC to test it specifically. |
| 5 | Skipping the Mock PDR | Schedule the mock 3 days before. Fresh eyes catch problems you miss. |
| 6 | Not ordering long-lead components | Order PCBs and specialty parts NOW. They arrive in time for SDII. |

| Sprint | Most Common Mistake | Prevention |
|---|---|---|
| 7 | Skipping the Hazard Assessment | Required before fabrication. PA must approve. No shortcuts. |
| 8 | BOM missing fasteners and consumables | Walk through the CAD assembly. Every visible component needs a BOM line. |
| 9–10 | Big-bang integration | Test each subsystem individually before connecting one at a time. |
| 11 | Leaving all testing to Sprint 12 | Start V&V testing NOW. Every week of margin is one redesign cycle. |
| 12–13 | Documentation debt | Write report sections during the sprint the work is done, not at the end. |
| 14 | Incomplete VCRM at FDR | Every requirement must be dispositioned: Verified, Failed (with disposition), Waived, or Deferred. |

# 4 SDI Sprint 0: Project Expo, Team Formation, Client Kickoff

*Weeks 1–2 | Course Text: Chapters 1, 2*

**▶ Top Priorities This Sprint**

1. Attend the Project Expo and complete the Project Selection Survey.
2. Meet your team, PA, and client.
3. Prepare for and conduct the Client Kickoff Meeting.
4. Set up the team repository (Teams group, file structure).
5. Begin the Individual Design Log (a personal notebook—physical or digital—where you record design decisions, sketches, calculations, and meeting notes throughout both semesters).

**What should I do before the kickoff meeting?**

Research the client's organization and industry. Prepare a slide deck for project discovery (you will present these during the Sprint 1 client kickoff). Write 10 open-ended questions. Assign roles: facilitator, scribe, timekeeper. See Course Text Ch. 2 for the full kickoff workflow.

Do not start designing during Sprint 0. Your job this sprint is to *understand the problem*, not to solve it. Listen more than you talk at the kickoff meeting. The first idea is rarely the best idea.

Send a follow-up email to the client within 24 hours of the kickoff meeting summarizing key decisions, open questions, and next steps. Copy your PA. This establishes your professionalism from Day 1.

### 4.1 Deliverables

- What to Expect from SD Quiz (Canvas)
- Project Selection Survey (by 11:30 AM on the designated date)
- Prof. Dev. – IP Module, Licensure Module
- Email to client to introduce team with team slide

## 5 SDI Sprint 1: The Project Plan, Scrum Training, Client Meeting Prep

*Weeks 3–4 | Course Text: Chapters 7, 11, 13*

**▶ Top Priorities This Sprint**

1. Develop the Statement of Work (SOW) with scope, deliverables, schedule, budget, and roles.
2. Establish the Product Backlog and first Sprint Backlog.
3. Set up task tracking (Trello, Planner, or GitHub Projects).
4. Complete Scrum training modules on Canvas.
5. Create the Team Charter.

**How detailed should the SOW be?**

Detailed enough that all parties agree on what "done" looks like for every deliverable. Include explicit "Out of Scope" items. Your Project Advisor (PA) evaluates the project against the SOW. See Course Text Ch. 7 for the complete SOW structure.

The most common Sprint 1 mistake is skipping the "Out of Scope" section. If you don't document what you will *not* do, the client may assume it's included.

### 5.1 Deliverables

- Statement of Work (SOW)
- Team Charter
- Prof. Dev. – Intro to Scrum, Creating Team Culture Reading
- Client Kickoff Meeting slides and meeting minutes
- Initial Product Backlog
- EHS Safety Training (Canvas module)

## 6 SDI Sprint 2: Concept Generation, Trade Study, Career Day

*Weeks 5–6 | Course Text: Chapters 4, 20*

**▶ Top Priorities This Sprint**

1. Generate ≥15 concepts using brainstorming, SCAMPER, and morphological charts.
2. Screen concepts with Pugh matrix; advance top 3–5 to decision matrix.
3. Prepare for Career Day (resume, elevator pitch, professional attire).
4. Sprint 1 Review / Retrospective / Sprint 2 Backlog.

Career Day is a prime networking opportunity. Bring 10–15 printed resumes and practice your capstone elevator pitch. See Course Text Ch. 20 for Career Day preparation.

Do not settle on a concept before generating alternatives. Three variations of the same idea are *not* three concepts. Generate genuinely different approaches before converging.

### 6.1 Deliverables

- Concept generation evidence (sketches, morphological chart)
- Prof. Dev. – Adv. Scrum Training
- Sprint 1 Review / Retrospective documentation
- Project Plan emailed to client

## 7 SDI Sprint 3: Concept Development, Client Approval of Approach

*Weeks 7–8 | Course Text: Chapters 3, 4, 17*

**▶ Top Priorities This Sprint**

1. Complete the weighted decision matrix with sensitivity analysis.
2. Present top 2–3 concepts to client; obtain approval of selected approach.
3. Draft the SDRD with initial requirements.
4. Identify applicable engineering standards.
5. Sprint 2 Review / Retrospective / Sprint 3 Backlog.

**What if the client doesn't like any of our concepts?**

Ask which aspects they do like and which concern them. Use their feedback to generate a hybrid concept or explore a new direction. Client approval is a scope commitment; invest the time to get it right.

### 7.1 Deliverables

- Decision matrix with sensitivity analysis
- Client approval of proposed solution (documented in meeting minutes)
- Engineering Standards identification
- Ethics Assignment / Peer Evaluation #1

# 8 SDI Sprint 4: Functional Prototypes / Site Plans, Ethics Assignment

*Weeks 9–10 | Course Text: Chapters 5, 12, 22*

**▶ Top Priorities This Sprint**

1. Build proof-of-concept prototype or develop analytical model.
2. Complete the SDRD (all requirements with TAID and MoSCoW).
3. Apply CLO 5 (sustainability/societal impact) sustainability tool (LEED, ENVISION, DFE, or ESJ). See Course Text Ch. 5, Table "Choosing the Right CLO 5 Tool" for guidance on which tool fits your project.
4. Complete the FDVSE value assessment (Feasibility, Desirability, Viability, Sustainability, Ethical Responsibility). See Course Text Ch. 5, Section 5.2 for the framework and guiding questions.
5. Complete the Ethics Assignment.
6. Sprint 3 Review / Retrospective / Sprint 4 Backlog.

Build Track: the PoC must demonstrate at least one physical function. "We drew it in CAD" is not a PoC. Paper Track: the PoC must demonstrate computational goal attainment (e.g., converging simulation, validated sub-model).

If using InnoHub shops, start machine training *now*. Training slots fill up quickly and you cannot use equipment without completing the required training on the Labriola Canvas site.

## 8.1 Deliverables

- Proof-of-Concept demonstration (photos/video)
- Ethics Assignment (1000–1500 word argumentative essay)
- Complete SDRD draft

# 9 SDI Sprint 5: PDR Preparation, Mock PDR, PDR Execution

*Weeks 11–13 | Course Text: Chapters 6, 8, 18, 19*

**▶ Top Priorities This Sprint**

1. Finalize PDR Report (PDRt) and send to client in advance.
2. Conduct Mock PDR (practice presentation with PA or another team).
3. Deliver PDR presentation (PDRw) to PA, client, and TAs.
4. Populate the VCRM (plan: all entries "Open" with TAID assigned).
5. Sprint 4 Review / Retrospective / Sprint 5 Backlog.

**What does a strong PDR look like?**

Strong PDRs have: testable requirements, ≥3 distinct concepts, a decision matrix with sensitivity analysis, proof-of-concept evidence (not just claims), an FDVSE value proposition statement, a sustainability assessment, and a team that can answer questions about every section. Before your PDR, the entire team must read and understand Course Text Chapters 6 (Verification), 8 (PDR), and 5 (Sustainability and FDVSE). See Course Text Ch. 8 for the full checklist and common failure patterns.

Do not skip the Mock PDR. Teams that rehearse consistently outperform teams that present cold. Schedule the mock at least 3 days before the formal review.

### 9.1 Deliverables

- PDR Report (PDRt) submitted to PA and client
- PDR Presentation (PDRw) delivered
- VCRM plan
- Individual Performance Eval #1

## 10 SDI Sprint 6: Design Refinement, Next Steps, SDII Handoff

*Weeks 14–15 | Course Text: Chapters 8, 10*

**▶ Top Priorities This Sprint**

1. Incorporate PDR feedback: categorize, assign owners, complete.
2. Baseline the SDRD (freeze; all future changes under change control; see Course Text Ch. 11 for the change control process).
3. Begin procurement of long-lead-time components for SDII.
4. Prepare SDII transition documentation.
5. Send Next Steps Email to client.

Order long-lead-time components *now*. Do not wait until SDII Sprint 8. Custom PCBs take 2–4 weeks. Specialty sensors take 2–4 weeks. Every week you wait in Sprint 6 is a week less margin in SDII.

### 10.1 Deliverables

- PDRt Update (incorporating feedback)
- Next Steps Email to client
- Peer Evaluation #2
- End-of-Semester Course Survey
- Sprint 5 Review / Retrospective / Sprint 6 documentation

### 10.2 Preparing for SDII

The semester break between SDI and SDII is *not* dead time. Before leaving for winter break: ensure all long-lead-time orders are placed, finalize your SDII transition documentation, and confirm your client's availability for the first week of SDII (Sprint 7). Week numbering resets to 1 in SDII. Review the Sprint 7 section of this guide before the first day of classes so you are ready to hit the ground running.

## 11 SDII Sprint 7: Re-engagement, Goal Setting, Hazard Assessment

*Week 1 | Course Text: Chapters 2, 11, 21*

**▶ Top Priorities This Sprint**

1. Send Re-engagement Email to client (5 required elements: reintroduce team, summarize SDI status, propose SDII goals, request meeting, attach Group Reflection).
2. Complete the Hazard Assessment and Safety Plan.
3. Review and update the Product Backlog with client.
4. Set SDII semester goals with PA and client.

The Hazard Assessment and Safety Plan must be approved by your PA *before* fabrication begins. Do not start building without it. See Course Text Ch. 21 for the template and required content.

### 11.1 Deliverables

- Re-engagement Email and Group Reflection & Action Plan
- EHS Safety Training (SDII module)
- Hazard Assessment & Safety Plan

## 12 SDII Sprint 8: CDR Report, Eng Calc Package, CDR Execution

*Weeks 2–5 | Course Text: Chapters 6 (Verification), 9 (CDR), 14 (Budget), 15 (Purchasing)*

**▶ Top Priorities This Sprint**

1. Finalize CDR Report (CDRt) and Engineering Calculation Package.
2. Send CDRt to client before CDR presentation.
3. Deliver CDR presentation (CDRw) to PA, client, and TAs.
4. Complete BOM with all components ordered or received.
5. Complete InnoHub Manufacturing Review Meeting (required before ordering build materials).

The Manufacturing Review Meeting is a **required checkoff on the purchasing form**. CP will not process material orders until it is complete. Schedule it early; slots fill up around CDR time.

Freeze the design at CDR. After this point, changes require formal change control (see Course Text Ch. 11 for the change control process). Teams that keep redesigning after CDR arrive at FDR with no working prototype.

### 12.1 Deliverables

- CDR Report (CDRt) with Engineering Calculation Package
- CDR Presentation (CDRw)
- Peer Evaluation #1 (SDII)
- Next Steps Email (post-CDR) to client

## 13 SDII Sprints 9–11: Build/Test/Iterate, Broader Impacts Essay

*Weeks 6–11 | Course Text: Chapters 5, 6, 12*

**▶ Top Priorities This Sprint**

1. Fabricate, assemble, and wire your system (Build Track) or develop full-fidelity models (Paper Track).
2. Test subsystems individually *before* integrating (test each subsystem in isolation, then integrate one at a time).
3. Begin system-level integration testing.
4. Write the Broader Impacts Essay (SDII, 1000–1500 words argumentative).
5. Sprint Reviews / Retrospectives every 2 weeks.

**The Spring Break Shipping Trap**: campus is closed for a full week during Spring Break. If a component fails during testing and you need a replacement, order it *before* break. Plan all critical procurement to arrive at least one week before break starts.

Start subsystem testing as early as Sprint 9. Do *not* wait until Sprint 12 for your first test. Every week of testing margin you have is a week of margin for redesign if something fails.

### 13.1 Deliverables

- Broader Impacts Essay
- Mid-Semester Course Survey
- Peer Evaluation #2 (SDII) / Individual Performance Eval #1
- Sprint Review / Retrospective documentation (each sprint)

# 14 SDII Sprints 12–13: FDR Preparation, System V&V, Final Testing

*Weeks 12–14 | Course Text: Chapters 6, 10, 18*

**▶ Top Priorities This Sprint**

1. Complete all verification testing and update the VCRM.
2. Disposition any failed requirements (redesign, waiver, risk acceptance).
3. Write the FDR Report (FDRt) including as-designed vs. as-built comparison.
4. Write the O&M manual.
5. Design and submit the Showcase poster for printing (∼1 week before Showcase).

A VCRM with mostly "Open" entries at FDR indicates insufficient testing. Every requirement must be dispositioned: Verified, Failed (with disposition), Waived, or Deferred. See Course Text Ch. 6 for failure disposition options.

Write the O&M manual for a non-engineer. Use photographs, numbered steps, and clear language. The person who operates your system after you graduate may not have your technical background.

## 14.1 Deliverables

- FDR Report (FDRt) submitted to PA and client
- Complete VCRM (all entries dispositioned)
- O&M manual
- Showcase poster (submitted for printing)
- Design-Build-Test-Redesign Slide (a single slide summarizing key iterations: what was designed, built, tested, and redesigned throughout the project)

# 15 SDII Sprint 14: FDR, Showcase, Documentation, Final Checkout

*Weeks 15–16 | Course Text: Chapters 10, 19*

**▶ Top Priorities This Sprint**

1. Deliver FDR presentation (FDRw) to PA, client, and TAs.
2. Present at the Design Showcase (poster + live demonstration).
3. Complete all final deliverables.
4. Final Checkout with PA.

**What is Final Checkout?**

A closing meeting with your PA to: hand off all project materials, return borrowed equipment, submit the Digital Design Archive, review your Project Goal Attainment, and formally close the project. Schedule it before the last day of class.

### 15.1 Final Deliverables

- FDR Presentation (FDRw)
- Digital Design Archive (all project files—CAD, code, reports, test data, meeting minutes—organized in the required folder structure; see Course Text Ch. 10)
- "What I Wish I Would Have Known" team video
- Project Synopsis
- Next Steps Email (post-FDR) to client
- Project Goal Attainment (a brief summary demonstrating how the team met—or did not meet—each project goal established in the SOW)
- Peer Evaluation #3 / Individual Performance Eval #2
- Final Course Survey
- Final Checkout with PA

## 16 Design Review Preparation Checklist

*Reference | Course Text: Chapters 8, 9, 10*

Before every design review, complete the **Project Health Check** self-assessment (Course Text, Appendix B). Rate your team Green/Yellow/Red on each of the 12 failure patterns. Discuss any Yellow or Red ratings with your PA.

**1 week before:**

- Report draft complete and under team review.
- Slide deck outline complete with section owners.
- PoC / demo hardware tested and working.

**3 days before:**

- Report finalized and sent to client.
- Slides complete; first rehearsal conducted.
- Identified and researched likely Q&A questions.

**Day of:**

- Arrive 15 minutes early; test projector and demo equipment.
- Backup slides on USB drive.
- Assigned Q&A responders by topic.
- Scribe ready to record feedback.

## 17 What Your PA Expects

*Reference | Course Text: Chapters 11, 13, 17*

Your PA evaluates you based on both *what* you deliver and *how* you deliver it. Meeting the checklist is necessary but not sufficient.

**What separates an A team from a B team?**

**A teams**: communicate proactively (PA hears about problems before they become crises), incorporate feedback aggressively (PDR weaknesses are addressed by CDR), demonstrate engineering judgment (trade-offs are analyzed with data, not guesses), and produce professional-quality documentation (reports read like they were written by engineers, not by students rushing at midnight).
**B teams**: meet deadlines but do not exceed expectations, address feedback partially, make design decisions without data, and produce adequate but unpolished documentation.

The single most effective way to improve your PA's assessment: **close the feedback loop**. When your PA gives feedback at a sprint review, write it down, address it, and report back at the next meeting: "At the last review you asked about our thermal analysis. Here are the results." PAs notice when their feedback is taken seriously.

## 18 The Feedback Loop Survival Guide

*Reference | Course Text: Chapter 13*

Feedback Loop peer evaluations directly affect your individual grade. SDI has 2 Peer Evaluations and 1 Individual Performance Evaluation; SDII has 3 Peer Evaluations and 2 Individual Performance Evaluations. See Canvas for the specific schedule of each.

There are two types of individual assessment: **Peer Evaluations** (completed by your teammates about each other) and **Individual Performance Evaluations** (completed by your PA about you). Both inform your individual grade. Check Canvas for the specific schedule of each.

A pattern of low Feedback Loop scores (below 3.0 out of 5.0) combined with PA observation of non-participation can result in individual grade reduction, PIP initiation, or course failure regardless of team scores.

**How to receive good evaluations**: attend every meeting on time, complete your tasks by the agreed deadline, communicate proactively when you are behind, help teammates when you finish

early, participate actively in discussions (not just silently), and be responsive to messages within 24 hours.

**How to give useful evaluations**: be specific. "Jordan was great" does not help the PA calibrate. "Jordan completed the firmware ahead of schedule, debugged the I2C communication issue that was blocking the sensor team, and volunteered to write the CDR power analysis section when the original author was overwhelmed" gives the PA evidence to work with.

**Self-evaluation**: be honest. PAs cross-check your self-rating against peer ratings and their own observations. Over-rating yourself while peers rate you lower signals a lack of self-awareness. Acknowledging a specific weakness ("I was late on two deliverables in Sprint 10 due to competing course deadlines; I should have communicated the conflict earlier") signals maturity.

## 19 CFO and Purchasing Survival Guide

*Reference | Course Text: Chapters 14, 15*

The CFO role is one of the most time-consuming in SDII. Here is the essential workflow:

1. **Get PA approval** before *every* purchase.
2. Check the **approved vendor list** on Canvas.
3. Complete the **Manufacturing Review Meeting** before ordering build materials (required checkoff).
4. Submit the correct form to CP: Purchase Form, Shopping List, or Reimbursement Form.
5. Email CP (capstonepurchasing@mines.edu) with: team name/number, date, vendor, form used.
6. Update the **Budget Tracker** monthly and upload to Canvas by deadline.
7. Pick up packages promptly from Labriola 204.
8. Use the Mines **tax exemption certificate** for all purchases (taxes are not reimbursed).

Purchases made without PA approval may result in **personal financial liability**. Always get approval first.

## 20 Travel Planning Quick Reference

*Reference | Course Text: Chapter 16*

- PA must approve Go/No-Go with CLT before any spending.
- Prepare a draft budget before meeting with CP.
- Carpooling required; multiple drivers same trip not reimbursed.
- Mileage: Mines rate, from 1500 Illinois, Golden CO 80401. Save Google Maps PDF.
- Meal receipts: itemized, no alcohol (entire receipt rejected).
- Submit all receipts within one month of trip.

## 21 Resume and Career Day Preparation

*Reference | Course Text: Chapter 20*

- Update your resume with capstone contributions (quantify: budget managed, components designed, tests conducted).
- Practice your 60-second elevator pitch.
- Prepare 5–7 STAR stories from capstone for behavioral interviews.
- Dress business casual; bring 10–15 printed resumes.
- Connect with PA, client, and TA on LinkedIn after the project.

## 22 Module Checklist: Sprint-to-Chapter Mapping

| Sprint | Weeks | Primary Activities | Course Text Chapters |
|---|---|---|---|
| **SDI (EDNS 491)** | | | |
| 0 | 1–2 | Project Expo, team formation, client kickoff | Ch. 1, 2 |
| 1 | 3–4 | Project Plan, Scrum training, product backlog | Ch. 7, 11, 13 |
| 2 | 5–6 | Concept generation, trade study, Career Day | Ch. 4, 20 |
| 3 | 7–8 | Concept development, client approval | Ch. 3, 4, 17 |
| 4 | 9–10 | Prototypes / site plans, ethics assignment | Ch. 5, 12, 22 |
| 5 | 11–13 | PDR preparation, mock PDR, PDR execution | Ch. 6, 8, 18, 19 |
| 6 | 14–15 | Design refinement, next steps, SDII handoff | Ch. 8, 10 |
| **SDII (EDNS 492)** | | | |
| 7 | 1 | Re-engagement, goal setting, safety plan | Ch. 2, 11, 21 |
| 8 | 2–5 | CDR report, eng. calc package, CDR execution | Ch. 6, 9, 14, 15 |
| 9–10 | 6–8 | Build/test/iterate, Broader Impacts essay | Ch. 5, 6, 12 |
| 11 | 9–11 | System integration, V&V execution | Ch. 6 |
| 12–13 | 12–14 | FDR preparation, final testing, documentation | Ch. 6, 10, 18 |
| 14 | 15–16 | FDR, Showcase, archive, final checkout | Ch. 10, 19 |

## 23 Quick Reference: Capstone Abbreviations

| Abbr. | Meaning | Abbr. | Meaning |
|---|---|---|---|
| BOM | Bill of Materials | O&M | Operations & Maintenance |
| CDR | Critical Design Review | PA | Project Advisor |
| CF | Course Faculty | PDR | Preliminary Design Review |
| CLO | Course Learning Outcome | PIP | Performance Improvement Plan |
| CLT | Capstone Leadership Team | PoC | Proof of Concept |
| CP | Capstone Purchasing | Prof. Dev. | Professional Development |
| EHS | Environmental Health & Safety | SDI | Senior Design I (EDNS 491) |
| ESJ | Engineering for Social Justice | | |
| FDR | Final Design Review | SDII | Senior Design II (EDNS 492) |
| FMEA | Failure Mode & Effects Analysis | SDRD | System Design Req. Document |
| ICD | Interface Control Document | SOW | Statement of Work |
| InnoHub | Labriola Innovation Hub | TAID | Test, Analysis, Inspection, Demo |
| MoSCoW | Must/Should/Could/Won't | VCRM | Verification Cross-Ref. Matrix |
| MVP | Minimum Viable Product | | |

## 24 Emergency Contacts and Key Resources

| Resource | Contact |
|---|---|
| Emergency | 911 |
| Mines Police (non-emergency) | 303-273-3333 |
| Capstone Program Director | Dr. Kristy Csavina, kcsavina@mines.edu |
| Capstone Purchasing | capstonepurchasing@mines.edu, Labriola 204 |
| InnoHub Operations | Julia Roos, jroos@mines.edu |
| InnoHub General | innohub@mines.edu |
| EHS | 303-273-3316, https://www.mines.edu/ehs/ |
| Writing Center | https://writing.mines.edu/, appointments at mines.mywconline.com |
| Labriola Info | https://labriola.mines.edu/ |

# A Assignment and Rubric Reference

This appendix consolidates the key assignments and rubrics for both EDNS 491 (Senior Design I) and EDNS 492 (Senior Design II). Use this as a quick reference for understanding expectations, point allocations, and grading criteria. **Canvas remains the authoritative source** for due dates, submission links, and any mid-semester updates to rubrics.

Rubric criteria summaries in this appendix are condensed for reference. Always review the full rubric on Canvas before submitting—some criteria contain additional detail not captured here.

## A.1 EDNS 491 — Senior Design I Assignments

### A.1.1 Sprint Cycle Assignments (Sprints 0–5)

Sprint Cycles form the backbone of your Scrum-based project management. Each two-to-three-week sprint follows the same structure:

1. **Sprint Planning** — Define sprint goals, populate the Sprint Backlog, assign tasks.
2. **Execution** — Daily stand-ups, design work, client engagement.
3. **Sprint Review** — Present progress to client and PA (slide deck required).
4. **Sprint Retrospective** — Team reflection on process improvements (*kaizen*).

**Submission (each sprint):** Upload (1) the Sprint Review slide deck and (2) the team retrospective within 24 hours of your Sprint Review meeting.

Sprint Cycles 0–5 are scored at 0 points individually but contribute to your *Team Assessment – Project Progress* grade (10% of course). Your PA evaluates sprint engagement holistically.

The sprint themes in SDI are:

- **Sprint 0** — Engaging with the Problem (client kickoff preparation and meeting)
- **Sprint 1** — Planning Through Ambiguity (90% complete Project Plan)
- **Sprint 2** — Trade Study & Concept Development (concept selection table)
- **Sprint 3** — Defining Your Concept (primary concept with client alignment)
- **Sprint 4** — From Concept to Preliminary Design (prototyping, feasibility evidence)
- **Sprint 5** — Preliminary Design Reviews (PDR presentation and report)

### A.1.2 Project Kickoff & Client Meeting

| | |
|---|---|
| **Points** | 10 (team) |
| **Deliverables** | Team Information Slide, Stakeholder Document, Sprint 0 slide deck, follow-up email to client |
| **Key expectations** | Prepare a 5-slide deck demonstrating early engagement; send follow-up email within 24 hrs; document next steps and first Sprint Review date |

Arrive at the client meeting with thoughtful, open-ended questions about the *problem*, not premature solutions. Print color copies of your Team Information Slide for the client.

### A.1.3 Project Plan (Statement of Work)

| | |
|---|---|
| **Points** | 100 (team) — initial draft is ungraded feedback; updated submission is graded |
| **Audience** | Client (primary), PA, Technical Advisors |
| **Key sections** | Project summary, scope of work, customer needs, design requirements (tabulated), stakeholders, background research, engineering standards, budget, material handoff, design process, project schedule (Gantt chart), team members, management tools, communication strategies |

Do **not** send the initial draft submission to the client. The initial submission is for PA feedback only. The updated submission is shared with the client via email.

### A.1.4 Preliminary Design Report (PD Report)

| | |
|---|---|
| **Points** | 100 (team) — initial submission for PA feedback; updated submission is graded |
| **Audience** | Client (stand-alone document) |
| **Key sections** | Executive summary, scope & background, design requirements, concept exploration (3+ concepts with diagrams), concept critique (sustainability/equity/risk tool), concept selection, functional/spatial basis, life cycle analysis, detailed drawings, budget, schedule |

Submit the initial PD Report to your client *before* the Preliminary Design Review so they can prepare questions. The updated version incorporates feedback from the review and is attached to the Next Steps Email.

### A.1.5 Preliminary Design Review (Presentation)

| | |
|---|---|
| **Points** | 100 (team + individual components) |
| **Format** | 15–20 min presentation + 15–20 min Q&A |
| **Audience** | Client, mentors, PA, faculty, students |
| **Requirements** | Use Capstone presentation template; every member speaks and is listed on their slides; business casual or formal attire required |

### A.1.6 Ethics Assignment

| | |
|---|---|
| **Points** | 100 (individual) |
| **Format** | 750–1500 word essay, PDF, Times New Roman or Arial 12pt |
| **Task** | Identify a specific ethical dilemma in your Capstone project; analyze using engineering ethical frameworks and relevant Code of Ethics; determine a professional course of action |
| **Key distinction** | Must present a genuine *dilemma* (competing values), not a generic safety statement |

**Required sections:** (1) The dilemma, (2) Stakeholder analysis, (3) Alternative courses of action, (4) Application of Canons, (5) Rationale & decision, (6) Measurement, (7) References (IEEE), (8) Optional Appendix A if AI was used.

AI may be used as a "simulator and debater" (stakeholder simulation, blind-spot checks, proofreading) but **not** to write the draft or generate citations. If AI is used, Appendix A (AI Transparency Statement) is mandatory.

### A.1.7 Engineering Standards

| | |
|---|---|
| **Points** | 100 (individual) |
| **Task** | Select a relevant standard/regulation module, summarize it, explain how it applies to your project, share findings with the team, and schedule a team discussion with your PA |
| **Deliverables** | Summary document uploaded to Canvas and to the team's shared folder |

### A.1.8 Project Goal Setting & Goal Attainment

**Goal Setting** (10 pts, team): Early-semester document stating end-of-semester goals agreed upon by client and PA, with a clear definition of done.

**Goal Attainment** (100 pts, team): End-of-semester document restating goals and providing evidence of achievement (images, videos, links to drawings, etc.). PA evaluates based on reasonable goal-setting and systematic progress through sprint reviews.

If your team does not have a functional prototype, preliminary site drawings, schematics, or equivalent by the end of SDI, the team will not receive an A regardless of other assignment scores.

#### A.1.9 Individual Performance Evaluations

| | |
|---|---|
| **Eval #1** | 100 pts, mid-semester |
| **Eval #2** | 110 pts, end-of-semester (includes semester reflection questions) |
| **Assessed by** | Project Advisor, informed by peer evaluations, sprint review observations, and contribution to deliverables |
| **Submission** | Upload a short document describing your technical and professional contributions with evidence (links to documents, deliverables) |

If your evaluation falls below 70%, your PA will issue a Performance Improvement Plan (PIP). Address it promptly—this is a professional development tool, not a punishment.

#### A.1.10 Next Steps Email

| | |
|---|---|
| **Points** | 10 (team) |
| **Audience** | Client (CC your PA and Technical Advisors) |
| **Key elements** | Thank client, summarize PDR feedback and how it was addressed (action items table), engineering recommendation for next semester, proposed budget, SDII schedule, attach updated PD Report |
| **Submission** | Paste email text or state the date sent |

### A.2 EDNS 492 — Senior Design II Assignments

#### A.2.1 Sprint Cycle Assignments (Sprints 7–14)

SDII continues the Scrum cadence with Sprints 7–14. The structure mirrors SDI (planning, execution, review, retrospective). Key sprint milestones in SDII:

- **Sprint 7** — Re-engagement with client and team; action plan for semester
- **Sprint 8** — Critical Design Review preparation and delivery
- **Sprint 9** — Post-CDR iteration; first peer evaluation
- **Sprints 10–11** — Build, test, redesign cycle; mid-semester evaluation
- **Sprint 12** — Showcase preparation; "What We Wish We Would Have Known" video
- **Sprint 13** — Final Design Review and Report
- **Sprint 14** — Design Showcase and final deliverables

#### A.2.2 Re-engagement Email & Activity

| | |
|---|---|
| **Email** | Professional re-engagement email to client at start of SDII |
| **Activity** | Group reflection on SDI and action plan for SDII improvement |

### A.2.3 Critical Design Report (CDR)

| | |
|---|---|
| **Points** | 100 (team) |
| **Key sections** | Design narrative, engineering analysis & calculation package (with TA sign-off), risk assessment (FMEA or risk matrix), test plans & procedures (V&V), technical drawings, technical writing |
| **Critical requirement** | All calculations must have sign-off from a Technical Advisor or faculty member |

The CDR is your "license to build." If your engineering analysis has gaps or lacks TA sign-off, you risk delays in moving to fabrication and testing.

### A.2.4 Critical Design Review (Presentation)

| | |
|---|---|
| **Points** | 100 (team + individual) |
| **Criteria** | Context (20%), Content (25%), Process (25%), Presentation (20%), Graphics (10%) |
| **Format** | Same structure as PDR; use Capstone presentation template; business attire required |

### A.2.5 Final Design Report (FDR)

| | |
|---|---|
| **Points** | 100 (team) |
| **Key sections** | Project overview, design approach, engineering analysis, final solution (referencing how design criteria were met or not), project management (budget, WBS, schedule), lessons learned, technical writing |
| **Purpose** | Stand-alone document enabling future engineers to continue the work |

### A.2.6 Final Design Review (Presentation)

| | |
|---|---|
| **Points** | 100 (team + individual) |
| **Criteria** | Context (20%), Content (25%), Process (25%), Presentation (20%), Graphics (10%) |
| **Note** | Must be completed *before* the Design Showcase |

### A.2.7 Broader Impacts Essay

| | |
|---|---|
| **Points** | 100 (individual) |
| **Format** | Argumentative essay, 1000–1500 words, use provided template |
| **Task** | Argue how an engineered system related to your project has positively or negatively impacted society, the environment, and/or the economy. Must take a clear thesis and defend it against opposing arguments. |
| **Key distinction** | This is an *argument*, not a report. Must include opposing viewpoints and rebuttals. |

**Required sections:** (1) Introduction & thesis, (2) Background, (3) Proponent arguments, (4) Opponent arguments, (5) Rebuttal, (6) Conclusion, (7) References (IEEE), (8) Optional Appendix A if AI was used.

The top 10 essays are sent to a panel of alumni and faculty judges. Winners are recognized at the Awards Breakfast and published in the Design Showcase program.

#### A.2.8 Design Showcase (Poster)

| | |
|---|---|
| **Poster Draft** | 25 pts (team) — submit well before printing for PA feedback |
| **Showcase** | 100 pts (team) — scored by external judges (alumni, industry, faculty) |
| **Poster specs** | 48 W × 36 H in PowerPoint; min 24pt font in text boxes; 28–32pt recommended |
| **Judging criteria** | Poster & display, project overview, design approach, design solution, next steps (each on 1–5 scale) |
| **Attendance** | Mandatory; business casual or professional recommended |

#### A.2.9 Project Synopsis

| | |
|---|---|
| **Points** | 10 (team) — complete/incomplete grading |
| **Purpose** | Short summary for the Design Showcase program booklet |
| **Requirement** | Professional, concise, complete, fully proofread |

#### A.2.10 Digital Design Archive

End-of-project submission of all project files, documentation, CAD files, code, and deliverables in an organized digital archive for the client and future teams.

#### A.2.11 "What We Wish We Would Have Known" Video

Team-produced video sharing advice and lessons learned for future Capstone teams. Shown to incoming SDI students.

#### A.2.12 Manufacturing Readiness Review

Teams working in the Labriola truck bays or Thorson Design Center must complete safety training, hazard assessments, and access contracts before beginning fabrication. EHS sign-off is required for any safety-related work, chemical usage, or painting.

### A.3 Rubric Reference

The following tables summarize the grading criteria for the most important assignments. CLO (Course Learning Outcome) rows that do not count toward scores are omitted. Point values shown are the maximum for each criterion.

#### A.3.1 Preliminary Design Report Rubric

| Criterion | Pts | Exceeds | Meets / Adequate | Marginal |
|---|---|---|---|---|
| Executive Summary | 3 | Succinct, highlights all key points | May omit components or lack conciseness | Limited info; must read full report |

| Criterion | Pts | Exceeds | Meets / Adequate | Marginal |
|---|---|---|---|---|
| Project Scope & Background | 5 | Thorough problem statement, existing solutions, standards | Generally clear but missing some depth | Insufficient background |
| Design Requirements | 5 | Tabulated with verification methods, types, numbering | Table present but incomplete | Minimal discussion |
| Concept Exploration | 40 | 3+ unique, feasible concepts with diagrams, functional/spatial basis, pros/cons | Fewer concepts or missing functional basis | One idea with little exploration |
| Concept Critique | 15 | Actionable design updates from appropriate sustainability/equity/risk tool | Tool applied but results not well summarized | Poor attempt or no tool applied |
| Concept Selection | 10 | Analysis-driven recommendation linked to constraints | Some reasoning but gaps in justification | Unsupported opinion |
| Detailed Drawings | 4 | Complete, professional set in appendices | Present but incomplete or unprofessional | Not complete nor professional |
| Budget | 3 | Discusses allocation, BoM, initial expenses | Present but lacking detail | Missing |
| Project Schedule | 3 | Updated Gantt chart referenced in text, milestones discussed | Present but not fully discussed | Only included, not discussed |
| Technical Writing | 10 | Clear, concise, one voice, proofread | Generally well written with some errors | Many errors; difficult to read |
| Formatting & Citations | 2 | IEEE citations, ToC, consistent formatting | Some formatting or citation issues | No citations; poorly formatted |

### A.3.2 Preliminary Design Review Presentation Rubric

| Criterion | Pts | Exceeds Mastery | Below Mastery |
|---|---|---|---|
| Design Considerations | 16 | Thorough problem definition; precise goals, constraints, requirements; strong justification of safety, standards, broader impacts | Vague problem/goals; unclear requirements; little attention to safety or standards |
| Proposed Solutions | 20 | Advanced engineering concepts with strong analysis and clear documentation; deep understanding of tradeoffs | Concepts applied incorrectly; minimal analysis; little awareness of tradeoffs |

| Criterion | Pts | Exceeds Mastery | Below Mastery |
|---|---|---|---|
| Engineering Process | 20 | Sophisticated concepts, in-depth analysis, comprehensive documentation | Underdeveloped concepts; weak analysis |
| Team Presentation | 30 | Exceptionally clear, well-organized, engaging; polished delivery | Lacks clarity or organization; unprepared delivery |
| Communication & Graphics | 10 | Original, compelling graphics; proper citations; minimal text | Ineffective graphics; excessive text; no citations |
| Professional Attire | 4 | All members in business casual/formal | Lack of professional attire detracts from credibility |

### A.3.3 Critical Design Report Rubric

| Criterion | Pts | Exceeds | Meets / Adequate | Marginal |
|---|---|---|---|---|
| Design Narrative | 10 | Clear method/logic; narrative guides reader; work concisely summarized | Present but lacking in some areas | Limited attempt; significant improvement needed |
| Engineering Analysis & Calcs | 20 | Exceptionally detailed, error-free, TA-approved calc package | Some gaps or questions unanswered | Incomplete or lacking TA sign-off |
| Risk Assessment | 15 | Detailed critique with risk tool (FMEA/matrix); clear mitigation plans linked to top risks | Tool present but surface-level analysis | Feels like afterthought; no clear narrative |
| Test Plans (V&V) | 10 | Complete, descriptive; non-team-member could execute tests | Adequate detail but needs improvement | Work needs significant improvement |
| Technical Drawings | 15 | Professional-quality, properly annotated, complete set | Present but formatting issues or incomplete | Limited attempt at professional quality |
| Technical Knowledge | 5 | Appropriate for completion level; clear justification of calculations and design decisions | Adequate but needs improvement | Significant improvement needed |
| Technical Writing | 15 | Clear, concise, one voice, proofread | Several errors affecting clarity | Many errors; not senior-level |

### A.3.4 Critical Design Review Presentation Rubric

| Criterion | Pts | Full Marks | Poor |
|---|---|---|---|
| Context | 20 | Succinctly defines problem; clear goals, constraints; addresses safety and broader impacts | 10 |
| Content | 25 | Appropriate engineering concepts, analysis, documentation; understands design tradeoffs | 13 |
| Process | 25 | Strong critical thinking in selecting design methods; prioritizes stakeholders | 13 |
| Presentation | 20 | Extremely well-organized and well-coordinated; creates excitement | 10 |
| Graphics | 10 | Original, compelling, effective; cites all external images | 5 |

### A.3.5 Final Design Report Rubric

| Criterion | Pts | Exceeds | Meets / Adequate | Marginal |
|---|---|---|---|---|
| Project Overview | 10 | Clear history and chronology; usable by future engineers | Generally clear but missing content | Limited documentation |
| Design Approach | 10 | Problem-specific approach with correctly applied tools; meets constraints | Basic approach but lacking clarity | Vague or incomplete methodologies |
| Engineering Analysis | 15 | Exceptionally detailed, error-free, no gaps; incorporates prior iteration feedback | Some gaps; questions unanswered | Not up to senior-level standard |
| Final Solution | 30 | Fully described; testing demonstrates criteria met; supporting documentation complete | Adequate but incomplete documentation | Solution does not meet criteria |
| Project Management | 5 | Accurate, up-to-date budget, WBS, schedule; clear path forward | Present but lacking detail | Substantial omissions |
| Lessons Learned | 5 | Thoughtful discussion of deviations and key takeaways | Adequate but could be deeper | Conflicts with PA observations |
| Technical Writing | 15 | Clear, concise, one voice | Several errors; some parts difficult to read | Many errors; not senior-level |
| Spelling & Grammar | 5 | Error-free | A few errors | Too many for senior writing |

| Criterion | Pts | Exceeds | Meets / Adequate | Marginal |
|---|---|---|---|---|
| Formatting & Citations | 5 | IEEE citations; consistent formatting throughout | Some citation/formatting issues | No citations |

### A.3.6 Final Design Review Presentation Rubric

Identical structure to the CDR Presentation Rubric: Context (20%), Content (25%), Process (25%), Presentation (20%), Graphics (10%). See Section A.3.4.

### A.3.7 Ethics Assignment Rubric

| Criterion | Pts | Full Marks | Inadequate |
|---|---|---|---|
| Intro to Ethical Consideration | 20 | Thoughtful introduction; clearly relates to project; explains why it is an ethical challenge | Limited relevance to project |
| Affected Stakeholders | 15 | Thoughtful consideration of parties, interests, responsibilities | Limited discussion |
| Alternative Courses of Action | 15 | Thoughtful alternatives with consequences to stakeholders | Limited discussion |
| Ethics & Liability Elaboration | 15 | Effectively uses Code of Ethics; discusses liability impacts | Minimal use of codes in context |
| Rationale for Chosen Path | 20 | Clear, concise rationale based on ethical frameworks and values | Limited discussion |
| Technical Writing | 10 | Clear, concise, no grammatical errors | Many errors; difficult to read |
| Citations & Formatting | 5 | IEEE format; all non-general-knowledge cited | No references to Codes of Ethics |

### A.3.8 Broader Impacts Essay Rubric

| Criterion | Pts | Excellent | Needs Improvement / Poor |
|---|---|---|---|
| Relevant | 25 | Addresses prompt in a controversial, insightful, or novel way | Does not understand intent of assignment |

| | | | |
|---|---|---|---|
| Persuasive | 25 | Inspiring, thought-provoking; strong evidence of engineering education | Not overly creative; arguments weak |
| Argumentative Essay Format | 10 | Well-organized; thesis, proponent/opponent arguments, strong conclusion | Reads more like informative essay or report |
| Well Written | 20 | Compact, efficient, concise; could be considered for publication | Rambling, narrative, or noisy writing |
| Spelling, Grammar, Punctuation | 10 | Practically no errors; strong evidence of proofreading | Pervasive errors; significant distraction |
| References & Citations | 10 | Thoroughly researched; IEEE format; scholarly sources | Little to no evidence of research |

### A.3.9 Project Plan (Statement of Work) Rubric

| **Criterion** | **Pts** | **Exceeds** | **Marginal / No Evidence** |
|---|---|---|---|
| Project Plan Summary | 2 | Professional, complete, concise | Missing |
| Scope of Work | 5 | Clear, succinct problem statement; constraints well defined | Could be much improved |
| Customer Needs | 5 | Clear, detailed list with importance ratings | Not provided |
| Design Requirements | 10 | Tabulated (not bulleted); complete; linked to client needs | Incomplete and vague |
| Stakeholders | 3 | Clear articulation with roles and timing | Insufficient detail |
| Budget | 2 | Discusses allocation and fundraising plans | Not addressed |
| Background Research | 10 | Demonstrates excellent understanding; existing solutions discussed | No evidence of understanding |
| Eng. Standards | 9 | Identifies standards/regulations with detail on project applicability | One or no standard identified |
| Material Handoff | 3 | Solid list of deliverables to client | Limited |

| Criterion | Pts | Exceeds | Marginal / No Evidence |
|---|---|---|---|
| Performance Criteria | 3 | Thoughtful consideration of V&V approach | Limited attempt |
| Design Process | 5 | Thoughtful discussion of engineering design steps | Limited attempt |
| Project Schedule | 5 | Detailed Gantt chart with discussion | Only reports/reviews; missing milestones |
| Team Members | 3 | Table with names, skills, roles, contact info, advisors | Poorly constructed |
| Management Tools | 3 | Strong management plan articulated | Not addressed |
| Communication | 3 | Frequency, mediums, status report strategy | Missing information |
| Technical Writing | 20 | Clear, concise, one voice, proofread | Lacks clarity; no attempt to edit |
| Formatting & Citations | 2 | IEEE citations; consistent formatting | No citations; poorly formatted |

### A.3.10 Design Showcase Poster Rubric

| Criterion | Pts | Excellent | Poor |
|---|---|---|---|
| Tables, Graphs, Figures | 5 | Correctly labeled, titled, explained; visuals convey information | Not labeled; visuals do not convey info |
| Grammar & Spelling | 5 | Free of all errors | Significant errors; poor editing |
| Quality of Poster | 5 | Legible from 5+ ft; appropriate color/font; phrases not sentences | Not legible; poor color/font; text-heavy |
| Design Process / Terminology | 5 | Clear presentation of process and final selection; appropriate terminology defined | No design process presented |
| Logical Organization | 5 | Clear language and images in logical sequence; includes analysis, intro, conclusion, future work | Illogical sequence; little relevant info |

### A.3.11 Individual Performance Evaluation Rubric

| Criterion | Pts | Excellent | Poor / Unsatisfactory |
|---|---|---|---|
| Technical Contribution | 40 | Produces excellent work; goes above and beyond; seeks help when appropriate; submits early for review | Poor quality; needs to be assigned work; leaves tasks for others; work often late |
| Professional Contribution | 40 | Positive influence; completes all sprint tasks; helps others; demonstrates leadership; always on time; professional with all stakeholders | Unprofessional at times; does not complete sprint tasks; consistently late; unavailable outside class |
| Assignment Submission | 20 | Fully addresses technical and professional contributions with clear supporting evidence | Minimally addresses contributions with no evidence |

### A.3.12 Next Steps Email Rubric (SDI)

| Criterion | Pts | Full Marks |
|---|---|---|
| PDR Action Items | 10 | Clear, complete list with status |
| Project Budget & Plan | 10 | Complete budget provided; original plan referenced; updates discussed |
| Next Steps Discussion | 10 | Clear path forward; client expectations addressed; concerns noted |
| Technical Writing | 10 | Clear, concise, one voice |
| Spelling & Grammar | 5 | No errors |
| Formatting & Citations | 5 | IEEE format; consistent formatting |

## A.4 Grade Weights

### A.4.1 EDNS 491 — Senior Design I

| Assignment Group | Weight |
|---|---|
| Team Assignments — Design Reports | 35% |
| Individual Performance Evaluation | 20% |
| Individual Writing Assignments | 10% |
| Team Assessment — Project Progress | 10% |
| Team Assignments — Design Reviews | 10% |
| Prof. Dev. — Engineering Standards | 5% |
| Prof. Dev. — New Employee Training (Complete All) | 3% |
| Team — Design Modules | 3% |
| EHS Safety Training & Access to Labriola Spaces | 2% |
| Prof. Dev. — Peer Evals, Surveys, Reflections | 2% |
| **Total** | **100%** |

### A.4.2 EDNS 492 — Senior Design II

| Assignment Group | Weight |
|---|---|
| Performance Evaluations — Individual | 25% |
| Project Planning & Completion — Team | 25% |
| Individual Writing Assignment | 10% |
| Design Reviews — Team | 10% |
| Critical Design Report — Team | 10% |
| Capstone Design Showcase — Team | 10% |
| Professional Development — Individual | 5% |
| Final Design Report — Team | 5% |
| Final Checkout | 0% |
| **Total** | **100%** |

In SDI, the biggest grade drivers are Design Reports (35%) and Individual Performance Evaluation (20%). In SDII, Performance Evaluations (25%) and Project Planning & Completion (25%) dominate. Prioritize accordingly.

*This guide is a companion to the EDNS 491/492 Course Text. For detailed methodology, technical reference, and worked examples, consult the Course Text chapters referenced in each section above.*